LI/LH Multivariable & Vector Analysis (2MVA(3))

Problems for week 10

(Questions on this problem sheet are NOT assessed)

Problem 1. Let C be the unit circle in the plane, oriented counterclockwise, and consider the vector field $\boldsymbol{F} = x^2\hat{\imath} + xy\hat{\jmath}$. Find the flux of $\boldsymbol{F}$ through C by directly evaluating the line integral $\oint_C \boldsymbol{F} \cdot \hat{\mathrm{n}}\, \mathrm{d}s$.

Problem 2. Find a parametric representation for the following surfaces:

(a) The part of the sphere $x^2 + y^2 + z^2 = 4$ that lies above the cone $z = \sqrt{x^2 + y^2}$;

(b) The part of the cylinder $x^2 + z^2 = 9$ that lies above the xy-plane and between the planes $y = -4$ and $y = 4$.

Problem 3. Find the area of the surface S, where S is the part of the surface $z = 4-2x^2+y$ that lies above the triangle with vertices $(0, 0, 0), (1, 0, 0)$ and $(1, 1, 0)$.

Problem 4. Evaluate the following surface integrals:

(a)

$$\iint_S y^2 \, \mathrm{d}S,$$

where S is the part of the sphere $x^2 + y^2 + z^2 = 1$ that lies above the cone $z = \sqrt{x^2 + y^2}$.

(b)

$$\iint_S (x + y + z) \, \mathrm{d}S,$$

where S is the part of the half-cylinder $x^2 + z^2 = 1, z \geq 0$, that lies between the planes $y = 0$ and $y = 2$.

Problem 5. Evaluate the surface integral

$$\iint_S \boldsymbol{F} \cdot \mathrm{d}\mathbf{S}$$

for the given vector field $\boldsymbol{F}$ and oriented surface S. In other words, find the flux of $\boldsymbol{F}$ through S.

(a) $\boldsymbol{F}(x, y, z) = xy\,\hat{\imath} + yz\,\hat{\jmath} + zx\,\hat{\mathrm{k}}$, S is the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the square $0 \leq x \leq 1, 0 \leq y \leq 1$, and has upward orientation.

(b) $\boldsymbol{F}(x, y, z) = -x\,\hat{\imath} - y\,\hat{\jmath} + z^3\,\hat{\mathrm{k}}$, S is the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 1$ and $z = 3$ with downward orientation.

(c) $\boldsymbol{F}(x, y, z) = y\,\hat{\imath} - x\,\hat{\jmath} + 2z\,\hat{\mathrm{k}}$, S is the hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$, oriented downward.